

## 5. Class Based Modeling

Class-based modeling identifies the static structure of the NIMC Integrated Management System. Candidate nouns/noun phrases are screened through General Classification, then evaluated by Selection Criteria. The retained classes are refined into attributes, methods, CRC cards, and four module-level collaboration diagrams.

### 5.1 List of Nouns / Noun Phrases

1. NIMC account
2. participant
3. NIMC employee
4. name
5. company
6. designation
7. phone number
8. email address
9. NID number
10. email verification
11. password
12. password reset
13. existing identity
14. Classroom service
15. Library service
16. Dormitory service
17. training course
18. seminar
19. workshop
20. speaker
21. Course Coordination Team
22. Course Coordinator
23. classroom
24. classroom code
25. session plan
26. session topic
27. date
28. time
29. assigned speaker
30. learning resource
31. document
32. presentation
33. video
34. notes
35. class recording
36. attendance
37. overall attendance
38. engagement indicator
39. notification
40. assignment
41. examination
42. assessment result
43. submission
44. AI-based analysis
45. similarity indicator
46. plagiarism indicator
47. AI-generated-content indicator
48. speaker evaluation
49. rating
50. comment
51. course evaluation
52. course feedback
53. certificate
54. QR code
55. LLM-based assistant
56. library membership
57. visitor
58. assistant librarian
59. librarian
60. membership validity
61. digital membership card
62. membership ID

63. borrowing eligibility
64. borrowing limit
65. physical book
66. library catalogue
67. ISBN
68. category
69. publisher
70. issue date
71. due date
72. return date
73. overdue status
74. penalty
75. circulation information
76. e-library
77. digital resource
78. magazine
79. open-source book
80. access permission
81. download permission
82. usage statistics
83. LibraryAssistantAI
84. book review
85. book request
86. monthly library report
87. dormitory application
88. accommodation type
89. expected stay
90. single seat
91. double seat
92. Hostel Manager
93. approval link
94. Director General
95. Additional Director General
96. Director
97. room
98. seat
99. room capacity
100. occupied seats
101. available seats
102. dormitory fee
103. waiver
104. payment confirmation
105. room allocation
106. check-in
107. stay status
108. stay-extension request
109. cancellation request
110. dormitory issue
111. Hostel Super
112. Warden
113. issue status
114. check-out
115. dormitory dashboard
116. monthly dormitory report
117. Hostel AI
118. 3D map location

## **5.2 List of Nouns / Noun Phrases in Solution Space**

1. User
2. NIMC Account
3. Email Service
4. SMS Service
5. AI Service
6. Payment System
7. ClassroomParticipant
8. Speaker
9. Course Coordinator
10. Course Coordination Team
11. Course
12. NIMC Classroom

13. Course Enrollment
14. Course Session
15. Learning Resource
16. Attendance Record
17. Assessment
18. Submission
19. Submission Analysis Service
20. Evaluation
21. Certificate
22. Classroom AI Assistant
23. Library Member
24. Librarian
25. Library Resource
26. Physical Book
27. Digital Resource
28. E-Library
29. Borrowing Record
30. Book Review
31. Book Request
32. Library Assistant AI
33. Dorm User
34. Dormitory Application
35. Dormitory Staff
36. Hostel Manager
37. Approval Authority
38. Approval Record
39. Room
40. Seat
41. Room Allocation
42. Stay Record
43. Stay Change Request
44. Dormitory Issue
45. Hostel AI

### 5.3 List of Verbs / Verb Phrases

|          |          |           |
|----------|----------|-----------|
| open     | activate | verify    |
| sign in  | update   | request   |
| reset    | create   | share     |
| enter    | remove   | contact   |
| assign   | prepare  | change    |
| view     | upload   | access    |
| search   | sort     | filter    |
| store    | record   | calculate |
| generate | identify | notify    |
| submit   | evaluate | analyze   |
| flag     | review   | determine |
| rate     | collect  | manage    |

|           |           |          |
|-----------|-----------|----------|
| issue     | scan      | check    |
| borrow    | extend    | display  |
| select    | return    | apply    |
| override  | read      | download |
| recommend | moderate  | add      |
| maintain  | approve   | reject   |
| forward   | allocate  | confirm  |
| check in  | check out | report   |
| resolve   | release   | track    |
| configure | retrieve  | preserve |

## 5.4 General Classification

General Classification uses the following seven categories: 1. External Entity, 2. Things, 3. Occurrence / Event, 4. Role, 5. Organizational Unit / Subsystem, 6. Places, 7. Structures. A noun/noun phrase passes if it has classification 1 or at least three separate classification numbers.

| Noun / Noun Phrase          | GC Passed In |   |   | Comment                     |
|-----------------------------|--------------|---|---|-----------------------------|
| User                        | 4            | 5 | 7 | Passed - 3+ classifications |
| NIMC Account                | 2            | 5 | 7 | Passed - 3+ classifications |
| Email Service               | 1            | 5 | 7 | Passed - External Entity    |
| SMS Service                 | 1            | 5 | 7 | Passed - External Entity    |
| AI Service                  | 1            | 5 | 7 | Passed - External Entity    |
| Payment System              | 1            | 5 | 7 | Passed - External Entity    |
| ClassroomParticipant        | 1            | 4 | 7 | Passed - External Entity    |
| Speaker                     | 1            | 4 | 7 | Passed - External Entity    |
| Course Coordinator          | 1            | 4 | 5 | Passed - External Entity    |
| Course Coordination Team    | 1            | 4 | 5 | Passed - External Entity    |
| Course                      | 2            | 5 | 7 | Passed - 3+ classifications |
| NIMC Classroom              | 2            | 5 | 7 | Passed - 3+ classifications |
| Course Enrollment           | 2            | 3 | 7 | Passed - 3+ classifications |
| Course Session              | 2            | 3 | 7 | Passed - 3+ classifications |
| Learning Resource           | 2            | 5 | 7 | Passed - 3+ classifications |
| Attendance Record           | 2            | 3 | 7 | Passed - 3+ classifications |
| Assessment                  | 2            | 3 | 7 | Passed - 3+ classifications |
| Submission                  | 2            | 3 | 7 | Passed - 3+ classifications |
| Submission Analysis Service | 1            | 5 | 7 | Passed - External Entity    |
| Evaluation                  | 2            | 3 | 7 | Passed - 3+ classifications |
| Certificate                 | 2            | 3 | 7 | Passed - 3+ classifications |

| Noun / Noun Phrase     | GC Passed In |   |   | Comment                     |                          |
|------------------------|--------------|---|---|-----------------------------|--------------------------|
| Classroom AI Assistant | 2            | 5 | 7 | Passed - 3+ classifications |                          |
| Library Member         | 1            | 4 | 7 | Passed - External Entity    |                          |
| Librarian              | 1            | 4 | 5 | Passed - External Entity    |                          |
| Library Resource       | 2            | 5 | 7 | Passed - 3+ classifications |                          |
| Physical Book          | 1            | 2 | 7 | Passed - External Entity    |                          |
| Digital Resource       | 2            | 5 | 7 | Passed - 3+ classifications |                          |
| E-Library              | 2            | 5 | 7 | Passed - 3+ classifications |                          |
| Borrowing Record       | 2            | 3 | 7 | Passed - 3+ classifications |                          |
| Book Review            | 2            | 3 | 7 | Passed - 3+ classifications |                          |
| Book Request           | 2            | 3 | 7 | Passed - 3+ classifications |                          |
| Library Assistant AI   | 2            | 5 | 7 | Passed - 3+ classifications |                          |
| Dorm User              | 1            | 4 | 7 | Passed - External Entity    |                          |
| Dormitory Application  | 2            | 3 | 7 | Passed - 3+ classifications |                          |
| Dormitory Staff        | 1            | 4 | 5 | Passed - External Entity    |                          |
| Hostel Manager         | 1            | 4 | 5 | Passed - External Entity    |                          |
| Approval Authority     | 1            | 4 | 5 | Passed - External Entity    |                          |
| Approval Record        | 2            | 3 | 7 | Passed - 3+ classifications |                          |
| Room                   | 1            | 2 | 6 | 7                           | Passed - External Entity |
| Seat                   | 1            | 2 | 6 | 7                           | Passed - External Entity |
| Room Allocation        | 2            | 3 | 7 | Passed - 3+ classifications |                          |
| Stay Record            | 2            | 3 | 7 | Passed - 3+ classifications |                          |
| Stay Change Request    | 2            | 3 | 7 | Passed - 3+ classifications |                          |
| Dormitory Issue        | 2            | 3 | 7 | Passed - 3+ classifications |                          |
| Hostel AI              | 2            | 5 | 7 | Passed - 3+ classifications |                          |

## 5.5 Potential Class List

1. User
2. NIMC Account
3. Email Service
4. SMS Service
5. AI Service
6. Payment System
7. ClassroomParticipant
8. Speaker
9. Course Coordinator
10. Course Coordination Team
11. Course
12. NIMC Classroom
13. Course Enrollment
14. Course Session
15. Learning Resource
16. Attendance Record
17. Assessment
18. Submission
19. Submission Analysis Service
20. Evaluation
21. Certificate
22. Classroom AI Assistant
23. Library Member
24. Librarian
25. Library Resource
26. Physical Book

27. Digital Resource
28. E-Library
29. Borrowing Record
30. Book Review
31. Book Request
32. Library Assistant AI
33. Dorm User
34. Dormitory Application
35. Dormitory Staff
36. Hostel Manager
37. Approval Authority
38. Approval Record
39. Room
40. Seat
41. Room Allocation
42. Stay Record
43. Stay Change Request
44. Dormitory Issue
45. Hostel AI

## 5.6 Selection Criteria

Selection Criteria: 1. Retained Information, 2. Needed Service, 3. Multiple Attributes, 4. Common Attributes, 5. Common Operation, 6. Essential Requirements. A candidate passes if it satisfies at least three criteria or criterion 6. Passing a criterion does not prevent a class from being merged into a better retained class during refinement.

| Noun / Noun Phrase       | SC Passed - Reason                                                                                                                                                              | Comment                             |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| User                     | 1. Retained Information - userID, name<br>2. Needed Service - NIMC Account<br>3. Multiple Attributes - userID, name                                                             | Passed - retained                   |
| NIMC Account             | 1. Retained Information - accountID, userID<br>2. Needed Service - User<br>3. Multiple Attributes - accountID, userID                                                           | Passed - retained                   |
| Email Service            | 2. Needed Service - NIMC Account<br>3. Multiple Attributes - messageID, recipientEmail<br>6. Essential Requirement - Email verification and normal notifications are explicitly | Passed - retained (external)        |
| SMS Service              | 2. Needed Service - Course Coordinator<br>3. Multiple Attributes - messageID, recipientPhone<br>6. Essential Requirement - SMS for special notification cases                   | Passed - retained (external)        |
| AI Service               | 2. Needed Service - Assessment<br>3. Multiple Attributes - serviceID, providerName<br>6. Essential Requirement - Shared AI capability by analysis and assistant                 | Passed - retained (external)        |
| Payment System           | 1. Retained Information - transactionID, userID<br>2. Needed Service - Dormitory Application<br>3. Multiple Attributes - transactionID, userID                                  | Passed - retained (external)        |
| ClassroomParticipant     | 1. Retained Information - participantID, userID<br>2. Needed Service - User<br>3. Multiple Attributes - participantID, userID                                                   | Passed - retained                   |
| Speaker                  | 1. Retained Information - speakerID, name<br>2. Needed Service - Course Coordinator<br>3. Multiple Attributes - speakerID, name                                                 | Passed - retained                   |
| Course Coordinator       | 1. Retained Information - coordinatorID, userID<br>2. Needed Service - Course<br>3. Multiple Attributes - coordinatorID, userID                                                 | Passed - retained                   |
| Course Coordination Team | 1. Retained Information - teamID, courseID<br>2. Needed Service - Course Coordinator<br>3. Multiple Attributes - teamID, courseID                                               | Passed - merged: Course Coordinator |
| Course                   | 1. Retained Information - courseID, title<br>2. Needed Service - Course Coordinator<br>3. Multiple Attributes - courseID, title                                                 | Passed - retained                   |

| Noun / Noun Phrase          | SC Passed - Reason                                                                                                                                          | Comment                                                                      |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| NIMC Classroom              | 1. Retained Information - classroomID, courseID<br>2. Needed Service - Course<br>3. Multiple Attributes - classroomID, courseID                             | Passed - retained                                                            |
| Course Enrollment           | 1. Retained Information - enrollmentID, participantID<br>2. Needed Service - NIMC Classroom<br>3. Multiple Attributes - enrollmentID, participantID         | Passed - merged: ClassroomParticipant / NIMC Classroom                       |
| Course Session              | 1. Retained Information - sessionID, courseID<br>2. Needed Service - Course<br>3. Multiple Attributes - sessionID, courseID                                 | Passed - retained                                                            |
| Learning Resource           | 1. Retained Information - resourceID, courseID<br>2. Needed Service - Course Coordinator<br>3. Multiple Attributes - resourceID, courseID                   | Passed - retained                                                            |
| Attendance Record           | 1. Retained Information - attendanceID, participantID<br>2. Needed Service - Required service<br>3. Multiple Attributes - attendanceID, participantID       | Passed - merged: Course Session / ClassroomParticipant // Course Coordinator |
| Assessment                  | 1. Retained Information - assessmentID, courseID<br>2. Needed Service - ClassroomParticipant<br>3. Multiple Attributes - assessmentID, courseID             | Passed - retained                                                            |
| Submission                  | 1. Retained Information - submissionID, assessmentID<br>2. Needed Service - Assessment<br>3. Multiple Attributes - submissionID, assessmentID               | Passed - merged: Assessment                                                  |
| Submission Analysis Service | 2. Needed Service - Assessment<br>3. Multiple Attributes - analysisID, submissionID<br>6. Essential Requirement - AI-based submission analysis              | Passed - merged: Assessment using shared AI Service                          |
| Evaluation                  | 1. Retained Information - evaluationID, courseID<br>2. Needed Service - ClassroomParticipant<br>3. Multiple Attributes - evaluationID, courseID             | Passed - retained                                                            |
| Certificate                 | 1. Retained Information - certificateID, participantID<br>2. Needed Service - ClassroomParticipant<br>3. Multiple Attributes - certificateID, participantID | Passed - retained                                                            |
| Classroom AI Assistant      | 1. Retained Information - assistantID, courseID<br>2. Needed Service - ClassroomParticipant<br>3. Multiple Attributes - assistantID, courseID               | Passed - retained                                                            |
| Library Member              | 1. Retained Information - libraryMemberID, userID<br>2. Needed Service - User<br>3. Multiple Attributes - libraryMemberID, userID                           | Passed - retained                                                            |

| Noun / Noun Phrase    | SC Passed - Reason                                                                                                                                          | Comment                                                         |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| Librarian             | 1. Retained Information - librarianID, userID<br>2. Needed Service - Library Member<br>3. Multiple Attributes - librarianID, userID                         | Passed - retained; Assistant Librarian merged                   |
| Library Resource      | 1. Retained Information - resourceID, title<br>2. Needed Service - Library Member<br>3. Multiple Attributes - resourceID, title                             | Passed - retained                                               |
| Physical Book         | 1. Retained Information - resourceID, isbn<br>2. Needed Service - Required service<br>3. Multiple Attributes - resourceID, isbn                             | Passed - merged: Library Resource by resourceFormat             |
| Digital Resource      | 1. Retained Information - resourceID, fileLink<br>2. Needed Service - E-Library<br>3. Multiple Attributes - resourceID, fileLink                            | Passed - merged: Library Resource by resourceFormat             |
| E-Library             | 1. Retained Information - eLibraryID, digitalResourceIDs<br>2. Needed Service - Library Resource<br>3. Multiple Attributes - eLibraryID, digitalResourceIDs | Passed - retained                                               |
| Borrowing Record      | 1. Retained Information - borrowingID, memberID<br>2. Needed Service - Required service<br>3. Multiple Attributes - borrowingID, memberID                   | Passed - merged: Library Member / Library Resource // Librarian |
| Book Review           | 1. Retained Information - reviewID, memberID<br>2. Needed Service - Required service<br>3. Multiple Attributes - reviewID, memberID                         | Passed - merged: Library Resource / Library Member              |
| Book Request          | 1. Retained Information - requestID, memberID<br>2. Needed Service - Librarian<br>3. Multiple Attributes - requestID, memberID                              | Passed - merged: Library Resource / Library Member              |
| Library Assistant AI  | 1. Retained Information - assistantID, userQuery<br>2. Needed Service - Library Member<br>3. Multiple Attributes - assistantID, userQuery                   | Passed - retained                                               |
| Dorm User             | 1. Retained Information - dormUserID, userID<br>2. Needed Service - User<br>3. Multiple Attributes - dormUserID, userID                                     | Passed - retained                                               |
| Dormitory Application | 1. Retained Information - applicationID, dormUserID<br>2. Needed Service - Dorm User<br>3. Multiple Attributes - applicationID, dormUserID                  | Passed - retained                                               |
| Dormitory Staff       | 1. Retained Information - staffID, userID<br>2. Needed Service - Hostel Manager<br>3. Multiple Attributes - staffID, userID                                 | Passed - merged: Hostel Manager for the simplified model        |

| Noun / Noun Phrase  | SC Passed - Reason                                                                                                                                         | Comment                                                   |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| Hostel Manager      | 1. Retained Information - managerID, userID<br>2. Needed Service - Dormitory Application<br>3. Multiple Attributes - managerID, userID                     | Passed - retained                                         |
| Approval Authority  | 1. Retained Information - authorityID, userID<br>2. Needed Service - Dormitory Application<br>3. Multiple Attributes - authorityID, userID                 | Passed - retained                                         |
| Approval Record     | 1. Retained Information - approvalID, applicationID<br>2. Needed Service - Dormitory Application<br>3. Multiple Attributes - approvalID, applicationID     | Passed - merged: Dormitory Application as approvalHistory |
| Room                | 1. Retained Information - roomID, roomNumber<br>2. Needed Service - Seat<br>3. Multiple Attributes - roomID, roomNumber                                    | Passed - retained                                         |
| Seat                | 1. Retained Information - seatID, roomID<br>2. Needed Service - Room<br>3. Multiple Attributes - seatID, roomID                                            | Passed - retained                                         |
| Room Allocation     | 1. Retained Information - allocationID, applicationID<br>2. Needed Service - Dormitory Application<br>3. Multiple Attributes - allocationID, applicationID | Passed - retained                                         |
| Stay Record         | 1. Retained Information - stayID, allocationID<br>2. Needed Service - Required service<br>3. Multiple Attributes - stayID, allocationID                    | Passed - merged: Room Allocation / Dorm User              |
| Stay Change Request | 1. Retained Information - requestID, dormUserID<br>2. Needed Service - Dorm User<br>3. Multiple Attributes - requestID, dormUserID                         | Passed - represented in: Dorm User request                |
| Dormitory Issue     | 1. Retained Information - issueID, dormUserID<br>2. Needed Service - Required service<br>3. Multiple Attributes - issueID, dormUserID                      | Passed - represented in: Dorm User / Hostel Manager       |
| Hostel AI           | 1. Retained Information - assistantID, userQuery<br>2. Needed Service - Dorm User<br>3. Multiple Attributes - assistantID, userQuery                       | Passed - retained                                         |

## 5.7 Selected Classes

After assessment and merging, only the following 30 classes are retained.

### Shared / Common

1. User
2. NIMC Account
3. Email Service
4. SMS Service
5. AI Service
6. Payment System

### Classroom

7. ClassroomParticipant
8. Speaker
9. Course Coordinator
10. Course
11. NIMC Classroom
12. Course Session
13. Learning Resource
14. Assessment
15. Evaluation
16. Certificate
17. Classroom AI Assistant

### Library

18. Library Member
19. Librarian
20. Library Resource
21. E-Library
22. Library Assistant AI

### Dormitory

23. Dorm User
24. Dormitory Application
25. Hostel Manager
26. Approval Authority
27. Room
28. Seat
29. Room Allocation
30. Hostel AI

## 5.8 Operation / Method Selection

Verbs and verb phrases are evaluated as probable methods. A dash means the verb represents a user/interface action or general behavior and does not create a separate class method.

| Verb / Verb Phrase | Probable Method            | Class                |
|--------------------|----------------------------|----------------------|
| open               | -                          | -                    |
| activate           | activateExistingIdentity() | NIMC Account         |
| verify             | verifyEmail()              | NIMC Account         |
| sign in            | login()                    | NIMC Account         |
| update             | updateProfile()            | User                 |
| request            | requestStayExtension()     | Dorm User            |
| reset              | resetPassword()            | NIMC Account         |
| create             | createCourse()             | Course               |
| share              | publishUpdate()            | NIMC Classroom       |
| enter              | joinClassroom(code)        | ClassroomParticipant |
| remove             | removeParticipant()        | NIMC Classroom       |

| Verb / Verb Phrase | Probable Method               | Class                 |
|--------------------|-------------------------------|-----------------------|
| contact            | -                             | -                     |
| assign             | assignSpeaker()               | Course Coordinator    |
| prepare            | prepareSessionPlan()          | Course Coordinator    |
| change             | updateSession()               | Course Session        |
| view               | viewSessionPlan()             | ClassroomParticipant  |
| upload             | uploadResource()              | Course Coordinator    |
| access             | accessResource()              | ClassroomParticipant  |
| search             | searchResource()              | Learning Resource     |
| sort               | sortResources()               | Learning Resource     |
| filter             | filterResources()             | Learning Resource     |
| store              | -                             | -                     |
| record             | recordParticipantAttendance() | Course Session        |
| calculate          | calculateAverage()            | Evaluation            |
| generate           | generateCertificate()         | Certificate           |
| identify           | analyzeSubmission()           | Assessment            |
| notify             | sendNotification()            | Email Service         |
| submit             | submitWork()                  | Assessment            |
| evaluate           | submitSpeakerEvaluation()     | Evaluation            |
| analyze            | analyzeSubmission()           | Assessment            |
| flag               | -                             | -                     |
| review             | reviewFlags()                 | Course Coordinator    |
| determine          | calculateFee()                | Dormitory Application |
| rate               | submitSpeakerEvaluation()     | Evaluation            |
| collect            | -                             | -                     |
| manage             | manageCatalogue()             | Librarian             |
| issue              | issueResource()               | Librarian             |
| scan               | verifyQRCode()                | Certificate           |
| check              | checkEligibility()            | Certificate           |
| borrow             | issueResource()               | Librarian             |
| extend             | extendDueDate()               | Librarian             |
| display            | displayResources()            | NIMC Classroom        |
| select             | checkAvailability()           | Seat                  |
| return             | returnResource()              | Librarian             |
| apply              | applyWaiver()                 | Payment System        |
| override           | overridePenalty()             | Librarian             |
| read               | openResource()                | E-Library             |
| download           | downloadResource()            | E-Library             |
| recommend          | recommendResources()          | Library Assistant AI  |
| moderate           | moderateReview()              | Librarian             |
| add                | addResource()                 | Library Resource      |
| maintain           | updateResource()              | Library Resource      |

| Verb / Verb Phrase | Probable Method             | Class                  |
|--------------------|-----------------------------|------------------------|
| approve            | approve()                   | Approval Authority     |
| reject             | reject()                    | Approval Authority     |
| forward            | forwardToNextAuthority()    | Approval Authority     |
| allocate           | allocate()                  | Room Allocation        |
| confirm            | confirmPayment()            | Payment System         |
| check in           | recordCheckIn()             | Hostel Manager         |
| check out          | recordCheckOut()            | Hostel Manager         |
| report             | reportIssue()               | Dorm User              |
| resolve            | manageReportedIssue()       | Hostel Manager         |
| release            | release()                   | Room Allocation        |
| track              | viewApplicationStatus()     | Dorm User              |
| configure          | configureCertificateRules() | Course                 |
| retrieve           | retrieveCourseResources()   | Classroom AI Assistant |
| preserve           | -                           | -                      |

## 5.9 Analysis

1. The final model contains 30 classes. The revision intentionally removes or merges the classes marked "cut" by the project team while preserving their required information or behavior inside retained classes.
2. Course Coordination Team is merged into Course Coordinator. The coordinator now carries both leadership and authorized team operations such as resource upload, attendance recording, assessment evaluation, AI-flag review, and evaluation management.
3. Course Enrollment is removed as an independent class. NIMC Classroom maintains classroom membership through participantIDs, while ClassroomClassroomParticipant stores the classrooms it has joined.
4. Attendance Record is removed as an independent class. Course Session stores per-session attendanceList, ClassroomParticipant keeps attendanceSummary, and Course Coordinator performs attendance recording.
5. Submission and Submission Analysis Service are removed as separate classes. Assessment stores submitted work, results, and analysis flags and calls the shared AI Service when analysis is required. Human review remains with the Course Coordinator.
6. Physical Book and Digital Resource are merged into Library Resource through resourceFormat and access/circulation attributes. The retained E-Library class provides the digital-access subsystem.
7. Borrowing Record is removed as a standalone class. Library Resource retains current borrower, issue/due/return dates, and circulationHistory; Library Member retains current borrowed resource IDs; Librarian performs issue and return operations.
8. Book Review and Book Request are removed as independent classes. Reviews, request demand, moderation, and user request/review actions are represented by Library Resource, Library Member, and Librarian.
9. Dormitory Staff is merged into Hostel Manager for this simplified model. The Hostel Manager therefore covers designated staff activities needed for check-in, check-out, occupancy management, and issue handling.
10. Approval Record is merged into Dormitory Application as approvalHistory and currentApprovalStage. Approval Authority represents DG, ADG, and Director using authorityRole and sequenceOrder.
11. Stay Record is merged into Room Allocation and Dorm User. Room Allocation maintains check-in, approved stay dates, checkout, and allocation status.
12. Stay Change Request and Dormitory Issue are not retained as classes. Dorm User exposes requestStayExtension(), requestCancellation(), and reportIssue(), while Hostel Manager manages the resulting decisions or actions.
13. Email Service and SMS Service remain external boundary classes because the usage story explicitly requires email verification/notifications and SMS in special cases. AI Service and Payment System similarly isolate shared external service interactions.
14. Classroom AI Assistant, Library Assistant AI, and Hostel AI remain distinct because each operates on different domain data and responsibilities, while AI Service provides the common underlying AI capability.

## 5.10 Selected Classes - Attributes and Methods

### 1. User

Attributes: userID, name, company, designation, phone, email, nidNumber, userType  
Methods: +viewProfile() +updateProfile() +activateIdentity() +accessModule()

### 2. NIMC Account

Attributes: accountID, userID, registeredEmail, passwordHash, emailVerified, accountStatus

- Methods: +registerAccount() +activateExistingIdentity() +verifyEmail() +login() +resetPassword() +updateCredentials()
- 3. Email Service**  
Attributes: messageID, recipientEmail, subject, messageType, deliveryStatus  
Methods: +sendVerificationEmail() +sendPasswordReset() +sendNotification() +sendApprovalLink()
- 4. SMS Service**  
Attributes: messageID, recipientPhone, messageType, deliveryStatus  
Methods: +sendSpecialNotification() +sendUrgentUpdate() +returnDeliveryStatus()
- 5. AI Service**  
Attributes: serviceID, providerName, modelName, requestID, serviceStatus  
Methods: +processPrompt() +analyzeContent() +generateResponse() +returnResult()
- 6. Payment System**  
Attributes: transactionID, userID, module, amount, paymentType, paymentStatus, waiverStatus  
Methods: +processPayment() +confirmPayment() +applyWaiver() +getPaymentStatus()
- 7. ClassroomParticipant**  
Attributes: participantID, userID, organizationName, designation, classroomIDs, attendanceSummary, engagementIndicator  
Methods: +joinClassroom(code) +viewSessionPlan() +accessResource() +viewAttendance() +submitAssessment() +evaluateSpeaker() +evaluateCourse() +viewCertificate()
- 8. Speaker**  
Attributes: speakerID, name, contactInformation, expertise, overallRating, ratingHistory  
Methods: +viewAssignedSessions() +provideMaterials() +viewOwnSchedule()
- 9. Course Coordinator**  
Attributes: coordinatorID, userID, courseID, teamMembers, accessStatus  
Methods: +createClassroom() +addParticipant() +removeParticipant() +assignSpeaker() +prepareSessionPlan() +updateSession() +uploadResource() +recordAttendance() +recordAssessmentResult() +reviewAIFlags() +manageEvaluation() +viewSpeakerRatingHistory() +issueCertificate()
- 10. Course**  
Attributes: courseID, title, courseType, startDate, endDate, description, certificateRules, status  
Methods: +createCourse() +updateCourse() +configureCertificateRules() +getSessions() +getParticipants() +changeStatus()
- 11. NIMC Classroom**  
Attributes: classroomID, courseID, classroomCode, participantIDs, resourceIDs, classroomStatus  
Methods: +generateCode() +admitParticipant() +removeParticipant() +listParticipants() +displayResources() +publishUpdate()
- 12. Course Session**  
Attributes: sessionID, courseID, topic, sessionDate, sessionTime, speakerID, mode, recordingLink, attendanceList, sessionStatus  
Methods: +scheduleSession() +updateSession() +assignSpeaker() +attachRecording() +recordParticipantAttendance() +notifyChange()
- 13. Learning Resource**  
Attributes: resourceID, courseID, sessionID, title, resourceType, fileLink, uploadDate, accessLevel  
Methods: +upload() +update() +searchResource() +sortResources() +filterResources() +openResource()
- 14. Assessment**  
Attributes: assessmentID, courseID, title, assessmentType, dueDate, submissionRecords, resultRecords, analysisFlags, status  
Methods: +createAssessment() +submitWork() +recordResult() +analyzeSubmission() +reviewFlags() +viewResult()
- 15. Evaluation**  
Attributes: evaluationID, courseID, sessionID, participantID, evaluationType, rating, comments, isAnonymous  
Methods: +submitSpeakerEvaluation() +submitCourseEvaluation() +calculateAverage() +viewSummary()
- 16. Certificate**  
Attributes: certificateID, participantID, courseID, issueDate, eligibilityStatus, qrCode  
Methods: +checkEligibility() +generateCertificate() +verifyQRCode() +viewCertificate()
- 17. Classroom AI Assistant**  
Attributes: assistantID, courseID, userQuery, retrievedResources, generatedResponse  
Methods: +askQuestion() +retrieveCourseResources() +answerFromResources() +returnSourceReference()
- 18. Library Member**  
Attributes: libraryMemberID, userID, category, membershipID, validFrom, validUntil, borrowingLimit, borrowedResourceIDs, qrCode, membershipStatus  
Methods: +activateMembership() +viewBorrowedResources() +viewDueDates() +searchCatalogue() +openDigitalResource() +requestBook() +submitReview() +askLibraryAI()
- 19. Librarian**  
Attributes: librarianID, userID, staffRole, permissions  
Methods: +verifyMember() +assignMembership() +manageCatalogue() +issueResource() +returnResource() +extendDueDate() +applyPenalty() +overridePenalty() +moderateReview() +reviewRequests() +generateMonthlyReport()
- 20. Library Resource**

Attributes: resourceID, title, author, isbn, category, publisher, publicationInfo, resourceFormat, availabilityStatus, accessPermission, downloadAllowed, currentBorrowerID, issueDate, dueDate, returnDate, circulationHistory, reviews, requestCount  
Methods: +addResource() +updateResource() +search() +markBorrowed() +markReturned() +setAccessPermission() +recordUsage() +addReview() +registerRequest()

#### 21. E-Library

Attributes: eLibraryID, digitalResourceIDs, publicAccessEnabled, usageStatistics  
Methods: +browse() +searchDigitalResources() +openResource() +downloadResource() +recordUsage()

#### 22. Library Assistant AI

Attributes: assistantID, userQuery, interestContext, courseContext, recommendedResources, generatedResponse  
Methods: +recommendResources() +getAvailability() +summarizeReviews() +suggestAlternative() +referToLibrarian()

#### 23. Dorm User

Attributes: dormUserID, userID, participantCategory, applicationID, allocatedRoomID, allocatedSeatID, stayStatus, reportedIssues  
Methods: +submitDormitoryApplication() +viewApplicationStatus() +viewAllocation() +requestStayExtension() +requestCancellation() +reportIssue() +viewIssueStatus() +viewStayDetails() +askHostelAI()

#### 24. Dormitory Application

Attributes: applicationID, dormUserID, nidNumber, preferredAccommodation, expectedStart, expectedEnd, seatType, reason, applicationStatus, currentApprovalStage, approvalHistory, feeAmount, waiverStatus, paymentStatus  
Methods: +submit() +updateStatus() +recordApproval() +calculateFee() +applyWaiver() +markPendingForAvailability() +confirmPaymentStatus()

#### 25. Hostel Manager

Attributes: managerID, userID, permissions, occupancySummary  
Methods: +reviewApplication() +checkAvailability() +allocateRoom() +recordCheckIn() +recordCheckOut() +manageOccupancy() +approveStayChange() +manageIssue() +generateMonthlyReport()

#### 26. Approval Authority

Attributes: authorityID, userID, authorityRole, sequenceOrder, decisionStatus  
Methods: +reviewApplication() +approve() +reject() +addRemark() +forwardToNextAuthority()

#### 27. Room

Attributes: roomID, roomNumber, roomType, totalCapacity, occupiedSeats, availableSeats, roomStatus  
Methods: +viewAvailability() +reserveSeat() +releaseSeat() +updateOccupancy()

#### 28. Seat

Attributes: seatID, roomID, seatNumber, seatType, availabilityStatus  
Methods: +checkAvailability() +allocate() +release()

#### 29. Room Allocation

Attributes: allocationID, applicationID, dormUserID, roomID, seatID, allocationDate, checkInDate, checkOutDate, approvedEndDate, allocationStatus  
Methods: +allocate() +activateStay() +updateStayDates() +recordCheckOut() +release() +viewDetails()

#### 30. Hostel AI

Attributes: assistantID, userQuery, roomContext, generatedResponse, actionStatus, mapReference  
Methods: +answerDormitoryQuestion() +retrieveRoomInfo() +provideNavigation() +controlSupportedDevice() +escalateToManager()

### 5.11 Class Cards

CRC cards use a standard monochrome format. The Collaborator column contains only class names. Collaboration lines are listed immediately after each completed card.

**Table CC.1: CRC Card for User**

| Class Name: User                                                                            |                                                                              |
|---------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| Attributes                                                                                  | Methods                                                                      |
| -userID<br>-name<br>-company<br>-designation<br>-phone<br>-email<br>-nidNumber<br>-userType | +viewProfile()<br>+updateProfile()<br>+activateIdentity()<br>+accessModule() |
| Responsibility                                                                              | Collaboration                                                                |
| Provide the shared identity and profile used to authenticate the person.                    | NIMC Account                                                                 |
| Provide the common user information required for the classroom role.                        | ClassroomParticipant                                                         |
| Provide the common user information required for library membership.                        | Library Member                                                               |
| Provide the common profile and required identity details for dormitory use.                 | Dorm User                                                                    |

**Collaborations:**

User --(opens / activates)--> NIMC Account  
 User --(provides profile to)--> ClassroomParticipant  
 User --(provides profile to)--> Library Member  
 User --(provides profile to)--> Dorm User

**Table CC.2: CRC Card for NIMC Account**

| <b>Class Name: NIMC Account</b>                                                                |                                                                                                                             |
|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>Attributes</b>                                                                              | <b>Methods</b>                                                                                                              |
| -accountID<br>-userID<br>-registeredEmail<br>-passwordHash<br>-emailVerified<br>-accountStatus | +registerAccount()<br>+activateExistingIdentity()<br>+verifyEmail()<br>+login()<br>+resetPassword()<br>+updateCredentials() |
| <b>Responsibility</b>                                                                          | <b>Collaboration</b>                                                                                                        |
| Link credentials to the shared person identity so duplicate identities are avoided.            | User                                                                                                                        |
| Use the registered email for verification and password-reset communication.                    | Email Service                                                                                                               |
| Provide the authenticated account used for authorized Classroom access.                        | ClassroomParticipant                                                                                                        |
| Provide the authenticated account used for authorized Library access.                          | Library Member                                                                                                              |
| Provide the authenticated account used for authorized Dormitory access.                        | Dorm User                                                                                                                   |

**Collaborations:**

NIMC Account --(authenticates)--> User  
 NIMC Account --(uses verification / reset through)--> Email Service  
 NIMC Account --(authorizes)--> ClassroomParticipant  
 NIMC Account --(authorizes)--> Library Member  
 NIMC Account --(authorizes)--> Dorm User

**Table CC.3: CRC Card for Email Service**

| <b>Class Name: Email Service</b>                                                         |                                                                                                |
|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| <b>Attributes</b>                                                                        | <b>Methods</b>                                                                                 |
| -messageID<br>-recipientEmail<br>-subject<br>-messageType<br>-deliveryStatus             | +sendVerificationEmail()<br>+sendPasswordReset()<br>+sendNotification()<br>+sendApprovalLink() |
| <b>Responsibility</b>                                                                    | <b>Collaboration</b>                                                                           |
| Send email-verification and password-reset messages for account access.                  | NIMC Account                                                                                   |
| Send normal classroom notices such as schedule, speaker, assessment, and course updates. | Course Coordinator                                                                             |
| Send library reminders and updates about due dates, requests, and resource availability. | Librarian                                                                                      |
| Send dormitory approval links and application/status notifications.                      | Dormitory Application                                                                          |

**Collaborations:**

Email Service --(verifies / resets for)--> NIMC Account  
Email Service --(sends classroom notices for)--> Course Coordinator  
Email Service --(sends library notices for)--> Librarian  
Email Service --(sends approval/status mail for)--> Dormitory Application

**Table CC.4: CRC Card for SMS Service**

| <b>Class Name: SMS Service</b>                                                |                                                                              |
|-------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| <b>Attributes</b>                                                             | <b>Methods</b>                                                               |
| -messageID<br>-recipientPhone<br>-messageType<br>-deliveryStatus              | +sendSpecialNotification()<br>+sendUrgentUpdate()<br>+returnDeliveryStatus() |
| <b>Responsibility</b>                                                         | <b>Collaboration</b>                                                         |
| Send special or urgent classroom updates when SMS is required.                | Course Coordinator                                                           |
| Send special library alerts when an additional notification method is needed. | Librarian                                                                    |
| Send urgent dormitory updates or exceptional notifications.                   | Hostel Manager                                                               |

**Collaborations:**

SMS Service --(sends urgent updates for)--> Course Coordinator  
SMS Service --(sends special alerts for)--> Librarian  
SMS Service --(sends urgent updates for)--> Hostel Manager

**Table CC.5: CRC Card for AI Service**

| <b>Class Name: AI Service</b>                                                               |                                                                                 |
|---------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| <b>Attributes</b>                                                                           | <b>Methods</b>                                                                  |
| -serviceID<br>-providerName<br>-modelName<br>-requestID<br>-serviceStatus                   | +processPrompt()<br>+analyzeContent()<br>+generateResponse()<br>+returnResult() |
| <b>Responsibility</b>                                                                       | <b>Collaboration</b>                                                            |
| Analyze assessment submissions and return similarity, plagiarism, or AI-content indicators. | Assessment                                                                      |
| Generate classroom answers from retrieved course material.                                  | Classroom AI Assistant                                                          |
| Generate library recommendations and responses from catalogue data.                         | Library Assistant AI                                                            |
| Generate dormitory assistance from authorized dormitory information.                        | Hostel AI                                                                       |

**Collaborations:**

AI Service --(analyzes submissions for)--> Assessment  
 AI Service --(generates responses for)--> Classroom AI Assistant  
 AI Service --(generates recommendations for)--> Library Assistant AI  
 AI Service --(generates responses for)--> Hostel AI

**Table CC.6: CRC Card for Payment System**

| <b>Class Name: Payment System</b>                                                                  |                                                                                 |
|----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| <b>Attributes</b>                                                                                  | <b>Methods</b>                                                                  |
| -transactionID<br>-userID<br>-module<br>-amount<br>-paymentType<br>-paymentStatus<br>-waiverStatus | +processPayment()<br>+confirmPayment()<br>+applyWaiver()<br>+getPaymentStatus() |
| <b>Responsibility</b>                                                                              | <b>Collaboration</b>                                                            |
| Process the applicable dormitory fee or approved waiver and return payment status.                 | Dormitory Application                                                           |
| Provide payment confirmation required before room allocation is completed.                         | Hostel Manager                                                                  |
| Provide the participant with the payment information/status related to the application.            | Dorm User                                                                       |

**Collaborations:**

Payment System --(processes fee / waiver for)--> Dormitory Application  
 Payment System --(confirms payment to)--> Hostel Manager  
 Payment System --(reports payment status to)--> Dorm User

**Table CC.7: CRC Card for ClassroomParticipant**

| <b>Class Name: ClassroomParticipant</b>                                                                                       |                                                                                                                                                                              |
|-------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Attributes</b>                                                                                                             | <b>Methods</b>                                                                                                                                                               |
| -participantID<br>-userID<br>-organizationName<br>-designation<br>-classroomIDs<br>-attendanceSummary<br>-engagementIndicator | +joinClassroom(code)<br>+viewSessionPlan()<br>+accessResource()<br>+viewAttendance()<br>+submitAssessment()<br>+evaluateSpeaker()<br>+evaluateCourse()<br>+viewCertificate() |
| <b>Responsibility</b>                                                                                                         | <b>Collaboration</b>                                                                                                                                                         |
| Use the shared NIMC identity as the participant profile.                                                                      | User                                                                                                                                                                         |
| Join the proper classroom using the classroom code and access the course space.                                               | NIMC Classroom                                                                                                                                                               |
| View session plans and personal session-attendance information.                                                               | Course Session                                                                                                                                                               |
| Search, filter, open, and use authorized course materials and recordings.                                                     | Learning Resource                                                                                                                                                            |
| Submit assignment/examination work and view recorded results.                                                                 | Assessment                                                                                                                                                                   |
| Submit speaker ratings and overall course feedback.                                                                           | Evaluation                                                                                                                                                                   |
| Receive and view the certificate when course eligibility requirements are met.                                                | Certificate                                                                                                                                                                  |

**Collaborations:**

```

ClassroomParticipant --(uses identity from)--> User
ClassroomParticipant --(joins)--> NIMC Classroom
ClassroomParticipant --(views schedule / attendance from)--> Course Session
ClassroomParticipant --(accesses)--> Learning Resource
ClassroomParticipant --(submits work to)--> Assessment
ClassroomParticipant --(submits feedback to)--> Evaluation
ClassroomParticipant --(receives)--> Certificate
    
```

**Table CC.8: CRC Card for Speaker**

| <b>Class Name: Speaker</b>                                                                   |                                                                      |
|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| <b>Attributes</b>                                                                            | <b>Methods</b>                                                       |
| -speakerID<br>-name<br>-contactInformation<br>-expertise<br>-overallRating<br>-ratingHistory | +viewAssignedSessions()<br>+provideMaterials()<br>+viewOwnSchedule() |
| <b>Responsibility</b>                                                                        | <b>Collaboration</b>                                                 |
| Receive course-session assignments and coordination information.                             | Course Coordinator                                                   |
| Conduct the assigned session and remain linked to its topic, date, and time.                 | Course Session                                                       |
| Provide documents, presentations, videos, notes, or other session materials.                 | Learning Resource                                                    |
| Be associated with participant ratings used to calculate speaker rating history.             | Evaluation                                                           |

**Collaborations:**

Speaker --(receives assignment from)--> Course Coordinator  
 Speaker --(conducts)--> Course Session  
 Speaker --(provides material to)--> Learning Resource  
 Speaker --(is rated through)--> Evaluation

**Table CC.9: CRC Card for Course Coordinator**

| <b>Class Name: Course Coordinator</b>                                        |                                                                                                                                                                                                                                                                                                            |
|------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Attributes</b>                                                            | <b>Methods</b>                                                                                                                                                                                                                                                                                             |
| -coordinatorID<br>-userID<br>-courseID<br>-teamMembers<br>-accessStatus      | +createClassroom()<br>+addParticipant()<br>+removeParticipant()<br>+assignSpeaker()<br>+prepareSessionPlan()<br>+updateSession()<br>+uploadResource()<br>+recordAttendance()<br>+recordAssessmentResult()<br>+reviewAIFlags()<br>+manageEvaluation()<br>+viewSpeakerRatingHistory()<br>+issueCertificate() |
| <b>Responsibility</b>                                                        | <b>Collaboration</b>                                                                                                                                                                                                                                                                                       |
| Create and manage the course and its configurable rules.                     | Course                                                                                                                                                                                                                                                                                                     |
| Create the classroom, manage participants, and control classroom access.     | NIMC Classroom                                                                                                                                                                                                                                                                                             |
| Contact and assign speakers to appropriate sessions.                         | Speaker                                                                                                                                                                                                                                                                                                    |
| Prepare/update the session plan and record participant attendance.           | Course Session                                                                                                                                                                                                                                                                                             |
| Upload and organize learning materials and class recordings.                 | Learning Resource                                                                                                                                                                                                                                                                                          |
| Create/manage assessments, record results, and review AI-generated flags.    | Assessment                                                                                                                                                                                                                                                                                                 |
| Manage speaker/course evaluation and view restricted speaker rating history. | Evaluation                                                                                                                                                                                                                                                                                                 |
| Check eligibility and coordinate issuing certificates to participants.       | Certificate                                                                                                                                                                                                                                                                                                |

**Collaborations:**

Course Coordinator --(manages)--> Course  
 Course Coordinator --(creates / controls)--> NIMC Classroom  
 Course Coordinator --(assigns)--> Speaker  
 Course Coordinator --(plans / updates)--> Course Session  
 Course Coordinator --(uploads / organizes)--> Learning Resource  
 Course Coordinator --(evaluates / reviews)--> Assessment  
 Course Coordinator --(manages)--> Evaluation  
 Course Coordinator --(issues)--> Certificate

**Table CC.10: CRC Card for Course**

| <b>Class Name: Course</b>                                                                                    |                                                                                                                               |
|--------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Attributes</b>                                                                                            | <b>Methods</b>                                                                                                                |
| -courseID<br>-title<br>-courseType<br>-startDate<br>-endDate<br>-description<br>-certificateRules<br>-status | +createCourse()<br>+updateCourse()<br>+configureCertificateRules()<br>+getSessions()<br>+getParticipants()<br>+changeStatus() |
| <b>Responsibility</b>                                                                                        | <b>Collaboration</b>                                                                                                          |
| Provide the main course information and rules managed by the coordinator.                                    | Course Coordinator                                                                                                            |
| Provide the course context represented by its NIMC classroom.                                                | NIMC Classroom                                                                                                                |
| Group and organize all planned sessions under the same course.                                               | Course Session                                                                                                                |
| Associate assignments/examinations and their results with the course.                                        | Assessment                                                                                                                    |
| Provide the configurable eligibility rules used for certificate generation.                                  | Certificate                                                                                                                   |

**Collaborations:**

Course --(is managed by)--> Course Coordinator  
 Course --(provides context to)--> NIMC Classroom  
 Course --(contains)--> Course Session  
 Course --(contains)--> Assessment  
 Course --(defines eligibility for)--> Certificate

**Table CC.11: CRC Card for NIMC Classroom**

| <b>Class Name: NIMC Classroom</b>                                                                  |                                                                                                                                  |
|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>Attributes</b>                                                                                  | <b>Methods</b>                                                                                                                   |
| -classroomID<br>-courseID<br>-classroomCode<br>-participantIDs<br>-resourceIDs<br>-classroomStatus | +generateCode()<br>+admitParticipant()<br>+removeParticipant()<br>+listParticipants()<br>+displayResources()<br>+publishUpdate() |
| <b>Responsibility</b>                                                                              | <b>Collaboration</b>                                                                                                             |
| Represent the online classroom belonging to a particular course.                                   | Course                                                                                                                           |
| Admit eligible participants using the classroom code and maintain the participant list.            | ClassroomParticipant                                                                                                             |
| Allow the coordinator to create, control, and update the classroom.                                | Course Coordinator                                                                                                               |
| Present the current session plan and session changes to classroom users.                           | Course Session                                                                                                                   |
| Present the course learning resources through one classroom platform.                              | Learning Resource                                                                                                                |

**Collaborations:**

NIMC Classroom --(belongs to)--> Course  
 NIMC Classroom --(admits)--> ClassroomParticipant  
 NIMC Classroom --(is controlled by)--> Course Coordinator  
 NIMC Classroom --(displays)--> Course Session  
 NIMC Classroom --(displays)--> Learning Resource

**Table CC.12: CRC Card for Course Session**

| <b>Class Name: Course Session</b>                                                                                                               |                                                                                                                                       |
|-------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>Attributes</b>                                                                                                                               | <b>Methods</b>                                                                                                                        |
| -sessionID<br>-courseID<br>-topic<br>-sessionDate<br>-sessionTime<br>-speakerID<br>-mode<br>-recordingLink<br>-attendanceList<br>-sessionStatus | +scheduleSession()<br>+updateSession()<br>+assignSpeaker()<br>+attachRecording()<br>+recordParticipantAttendance()<br>+notifyChange() |
| <b>Responsibility</b>                                                                                                                           | <b>Collaboration</b>                                                                                                                  |
| Belong to the relevant course and contribute to its overall session plan.                                                                       | Course                                                                                                                                |
| Link the assigned speaker to the session topic, date, and time.                                                                                 | Speaker                                                                                                                               |
| Provide participants with schedule information and store their session attendance.                                                              | ClassroomParticipant                                                                                                                  |
| Allow the coordinator to schedule, update, and record attendance for the session.                                                               | Course Coordinator                                                                                                                    |
| Trigger normal email notification when the session schedule or speaker changes.                                                                 | Email Service                                                                                                                         |
| Trigger SMS notification for special or urgent session changes.                                                                                 | SMS Service                                                                                                                           |

**Collaborations:**

Course Session --(belongs to)--> Course  
 Course Session --(is conducted by)--> Speaker  
 Course Session --(provides schedule / attendance to)--> ClassroomParticipant  
 Course Session --(is managed by)--> Course Coordinator  
 Course Session --(notifies through)--> Email Service  
 Course Session --(sends urgent notice through)--> SMS Service

**Table CC.13: CRC Card for Learning Resource**

| <b>Class Name: Learning Resource</b>                                                                          |                                                                                                          |
|---------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| <b>Attributes</b>                                                                                             | <b>Methods</b>                                                                                           |
| -resourceID<br>-courseID<br>-sessionID<br>-title<br>-resourceType<br>-fileLink<br>-uploadDate<br>-accessLevel | +upload()<br>+update()<br>+searchResource()<br>+sortResources()<br>+filterResources()<br>+openResource() |
| <b>Responsibility</b>                                                                                         | <b>Collaboration</b>                                                                                     |
| Allow the coordinator/team role to upload and maintain course material.                                       | Course Coordinator                                                                                       |
| Appear in the appropriate classroom for centralized resource access.                                          | NIMC Classroom                                                                                           |
| Link materials or recordings to the session in which they were used or delivered.                             | Course Session                                                                                           |
| Allow authorized participants to search, sort, filter, and access resources.                                  | ClassroomParticipant                                                                                     |
| Provide the approved knowledge source used to answer classroom questions.                                     | Classroom AI Assistant                                                                                   |

**Collaborations:**

Learning Resource --(is maintained by)--> Course Coordinator  
 Learning Resource --(is presented in)--> NIMC Classroom  
 Learning Resource --(is linked to)--> Course Session  
 Learning Resource --(is accessed by)--> ClassroomParticipant  
 Learning Resource --(provides knowledge to)--> Classroom AI Assistant

**Table CC.14: CRC Card for Assessment**

| <b>Class Name: Assessment</b>                                                                                                            |                                                                                                                    |
|------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| <b>Attributes</b>                                                                                                                        | <b>Methods</b>                                                                                                     |
| -assessmentID<br>-courseID<br>-title<br>-assessmentType<br>-dueDate<br>-submissionRecords<br>-resultRecords<br>-analysisFlags<br>-status | +createAssessment()<br>+submitWork()<br>+recordResult()<br>+analyzeSubmission()<br>+reviewFlags()<br>+viewResult() |
| <b>Responsibility</b>                                                                                                                    | <b>Collaboration</b>                                                                                               |
| Receive participant submissions and expose the participant result after evaluation.                                                      | ClassroomParticipant                                                                                               |
| Allow the coordinator/team role to create assessments, evaluate work, record results, and review flags.                                  | Course Coordinator                                                                                                 |
| Remain associated with the course whose performance and completion it measures.                                                          | Course                                                                                                             |
| Send supported submission content for AI analysis and receive non-final indicators.                                                      | AI Service                                                                                                         |

**Collaborations:**

Assessment --(receives submission from)--> ClassroomParticipant  
 Assessment --(is evaluated by)--> Course Coordinator  
 Assessment --(belongs to)--> Course  
 Assessment --(requests analysis from)--> AI Service

**Table CC.15: CRC Card for Evaluation**

| <b>Class Name: Evaluation</b>                                                                                         |                                                                                                  |
|-----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| <b>Attributes</b>                                                                                                     | <b>Methods</b>                                                                                   |
| -evaluationID<br>-courseID<br>-sessionID<br>-participantID<br>-evaluationType<br>-rating<br>-comments<br>-isAnonymous | +submitSpeakerEvaluation()<br>+submitCourseEvaluation()<br>+calculateAverage()<br>+viewSummary() |
| <b>Responsibility</b>                                                                                                 | <b>Collaboration</b>                                                                             |
| Collect numerical ratings, comments, and course feedback from participants.                                           | ClassroomParticipant                                                                             |
| Associate session evaluations with the relevant speaker and update aggregate ratings.                                 | Speaker                                                                                          |
| Collect overall evaluation information for the relevant course.                                                       | Course                                                                                           |
| Allow the coordinator to manage evaluations and view restricted speaker rating history.                               | Course Coordinator                                                                               |

**Collaborations:**

Evaluation --(receives feedback from)--> ClassroomParticipant  
 Evaluation --(evaluates)--> Speaker  
 Evaluation --(evaluates)--> Course  
 Evaluation --(is managed by)--> Course Coordinator

**Table CC.16: CRC Card for Certificate**

| <b>Class Name: Certificate</b>                                                               |                                                                                        |
|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| <b>Attributes</b>                                                                            | <b>Methods</b>                                                                         |
| -certificateID<br>-participantID<br>-courseID<br>-issueDate<br>-eligibilityStatus<br>-qrCode | +checkEligibility()<br>+generateCertificate()<br>+verifyQRCode()<br>+viewCertificate() |
| <b>Responsibility</b>                                                                        | <b>Collaboration</b>                                                                   |
| Issue and display the certificate to the eligible participant.                               | ClassroomParticipant                                                                   |
| Use the course-specific configured conditions when checking eligibility.                     | Course                                                                                 |
| Allow the coordinator to check eligibility and generate the QR-coded certificate.            | Course Coordinator                                                                     |

**Collaborations:**

Certificate --(is issued to)--> ClassroomParticipant  
 Certificate --(uses rules from)--> Course  
 Certificate --(is generated by)--> Course Coordinator

**Table CC.17: CRC Card for Classroom AI Assistant**

| <b>Class Name: Classroom AI Assistant</b>                                                 |                                                                                                    |
|-------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| <b>Attributes</b>                                                                         | <b>Methods</b>                                                                                     |
| -assistantID<br>-courseID<br>-userQuery<br>-retrievedResources<br>-generatedResponse      | +askQuestion()<br>+retrieveCourseResources()<br>+answerFromResources()<br>+returnSourceReference() |
| <b>Responsibility</b>                                                                     | <b>Collaboration</b>                                                                               |
| Receive participant questions and return course-related answers.                          | ClassroomParticipant                                                                               |
| Retrieve relevant documents, notes, presentations, videos, and recordings as evidence.    | Learning Resource                                                                                  |
| Use the common AI capability to generate a response grounded in retrieved course content. | AI Service                                                                                         |

**Collaborations:**

Classroom AI Assistant --(answers)--> ClassroomParticipant  
 Classroom AI Assistant --(retrieves from)--> Learning Resource  
 Classroom AI Assistant --(calls)--> AI Service

**Table CC.18: CRC Card for Library Member**

| <b>Class Name: Library Member</b>                                                                                                                                 |                                                                                                                                                                              |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Attributes</b>                                                                                                                                                 | <b>Methods</b>                                                                                                                                                               |
| -libraryMemberID<br>-userID<br>-category<br>-membershipID<br>-validFrom<br>-validUntil<br>-borrowingLimit<br>-borrowedResourceIDs<br>-qrCode<br>-membershipStatus | +activateMembership()<br>+viewBorrowedResources()<br>+viewDueDates()<br>+searchCatalogue()<br>+openDigitalResource()<br>+requestBook()<br>+submitReview()<br>+askLibraryAI() |
| <b>Responsibility</b>                                                                                                                                             | <b>Collaboration</b>                                                                                                                                                         |
| Use the shared NIMC identity as the basis for library membership.                                                                                                 | User                                                                                                                                                                         |
| Receive membership assignment/verification and borrowing service from library staff.                                                                              | Librarian                                                                                                                                                                    |
| Search, borrow/open, request, and review library resources as permitted.                                                                                          | Library Resource                                                                                                                                                             |
| Read or download digital resources according to their access permissions.                                                                                         | E-Library                                                                                                                                                                    |
| Ask for recommendations, availability, related resources, and review summaries.                                                                                   | Library Assistant AI                                                                                                                                                         |

**Collaborations:**

Library Member --(uses identity from)--> User  
 Library Member --(is verified / served by)--> Librarian  
 Library Member --(searches / borrows / reviews)--> Library Resource  
 Library Member --(accesses)--> E-Library  
 Library Member --(asks recommendations from)--> Library Assistant AI

**Table CC.19: CRC Card for Librarian**

| <b>Class Name: Librarian</b>                                                            |                                                                                                                                                                                                                                          |
|-----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Attributes</b>                                                                       | <b>Methods</b>                                                                                                                                                                                                                           |
| -librarianID<br>-userID<br>-staffRole<br>-permissions                                   | +verifyMember()<br>+assignMembership()<br>+manageCatalogue()<br>+issueResource()<br>+returnResource()<br>+extendDueDate()<br>+applyPenalty()<br>+overridePenalty()<br>+moderateReview()<br>+reviewRequests()<br>+generateMonthlyReport() |
| <b>Responsibility</b>                                                                   | <b>Collaboration</b>                                                                                                                                                                                                                     |
| Verify users, assign membership, and manage issuing/returning resources for members.    | Library Member                                                                                                                                                                                                                           |
| Maintain catalogue details, circulation status, penalties, reviews, and request demand. | Library Resource                                                                                                                                                                                                                         |
| Upload/manage digital resources, access permissions, and e-library usage information.   | E-Library                                                                                                                                                                                                                                |
| Send due-date reminders and other important library notifications.                      | Email Service                                                                                                                                                                                                                            |

**Collaborations:**

```

Librarian --(verifies / serves)--> Library Member
Librarian --(manages)--> Library Resource
Librarian --(manages)--> E-Library
Librarian --(sends notices through)--> Email Service

```

**Table CC.20: CRC Card for Library Resource**

| Class Name: Library Resource                                                                                                                                                                                                                                                                   |                                                                                                                                                                          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Attributes                                                                                                                                                                                                                                                                                     | Methods                                                                                                                                                                  |
| -resourceID<br>-title<br>-author<br>-isbn<br>-category<br>-publisher<br>-publicationInfo<br>-resourceFormat<br>-availabilityStatus<br>-accessPermission<br>-downloadAllowed<br>-currentBorrowerID<br>-issueDate<br>-dueDate<br>-returnDate<br>-circulationHistory<br>-reviews<br>-requestCount | +addResource()<br>+updateResource()<br>+search()<br>+markBorrowed()<br>+markReturned()<br>+setAccessPermission()<br>+recordUsage()<br>+addReview()<br>+registerRequest() |
| Responsibility                                                                                                                                                                                                                                                                                 | Collaboration                                                                                                                                                            |
| Provide searchable resource information and permitted borrowing/access functions to members.                                                                                                                                                                                                   | Library Member                                                                                                                                                           |
| Allow library staff to maintain metadata, availability, circulation, reviews, and request demand.                                                                                                                                                                                              | Librarian                                                                                                                                                                |
| Provide the digital form of a resource to the e-library when resourceFormat is digital.                                                                                                                                                                                                        | E-Library                                                                                                                                                                |
| Provide catalogue, availability, and review data for AI recommendations.                                                                                                                                                                                                                       | Library Assistant AI                                                                                                                                                     |

**Collaborations:**

```

Library Resource --(is used by)--> Library Member
Library Resource --(is managed by)--> Librarian
Library Resource --(supplies digital content to)--> E-Library
Library Resource --(supplies catalogue data to)--> Library Assistant AI
    
```

**Table CC.21: CRC Card for E-Library**

| <b>Class Name: E-Library</b>                                                             |                                                                                                    |
|------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| <b>Attributes</b>                                                                        | <b>Methods</b>                                                                                     |
| -eLibraryID<br>-digitalResourceIDs<br>-publicAccessEnabled<br>-usageStatistics           | +browse()<br>+searchDigitalResources()<br>+openResource()<br>+downloadResource()<br>+recordUsage() |
| <b>Responsibility</b>                                                                    | <b>Collaboration</b>                                                                               |
| Host and expose library resources that are available in digital form.                    | Library Resource                                                                                   |
| Allow members/public users to read or download resources according to access permission. | Library Member                                                                                     |
| Allow the librarian to manage digital access and review usage statistics.                | Librarian                                                                                          |

**Collaborations:**

E-Library --(hosts)--> Library Resource  
E-Library --(serves)--> Library Member  
E-Library --(is managed by)--> Librarian

**Table CC.22: CRC Card for Library Assistant AI**

| <b>Class Name: Library Assistant AI</b>                                                                         |                                                                                                                    |
|-----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| <b>Attributes</b>                                                                                               | <b>Methods</b>                                                                                                     |
| -assistantID<br>-userQuery<br>-interestContext<br>-courseContext<br>-recommendedResources<br>-generatedResponse | +recommendResources()<br>+getAvailability()<br>+summarizeReviews()<br>+suggestAlternative()<br>+referToLibrarian() |
| <b>Responsibility</b>                                                                                           | <b>Collaboration</b>                                                                                               |
| Receive the member's interests, course context, and recommendation questions.                                   | Library Member                                                                                                     |
| Search catalogue data, availability, reviews, and related resources when producing suggestions.                 | Library Resource                                                                                                   |
| Refer the user to library staff when a question requires human judgment or unavailable information.             | Librarian                                                                                                          |
| Use the common AI capability to generate recommendation and follow-up responses.                                | AI Service                                                                                                         |

**Collaborations:**

Library Assistant AI --(recommends to)--> Library Member  
 Library Assistant AI --(retrieves data from)--> Library Resource  
 Library Assistant AI --(refers special cases to)--> Librarian  
 Library Assistant AI --(calls)--> AI Service

**Table CC.23: CRC Card for Dorm User**

| <b>Class Name: Dorm User</b>                                                                                                               |                                                                                                                                                                                                                     |
|--------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Attributes</b>                                                                                                                          | <b>Methods</b>                                                                                                                                                                                                      |
| -dormUserID<br>-userID<br>-participantCategory<br>-applicationID<br>-allocatedRoomID<br>-allocatedSeatID<br>-stayStatus<br>-reportedIssues | +submitDormitoryApplication()<br>+viewApplicationStatus()<br>+viewAllocation()<br>+requestStayExtension()<br>+requestCancellation()<br>+reportIssue()<br>+viewIssueStatus()<br>+viewStayDetails()<br>+askHostelAI() |
| <b>Responsibility</b>                                                                                                                      | <b>Collaboration</b>                                                                                                                                                                                                |
| Use the shared NIMC profile and provide required identity information such as NID when needed.                                             | User                                                                                                                                                                                                                |
| Submit and track the dormitory application and its approval/payment status.                                                                | Dormitory Application                                                                                                                                                                                               |
| View the allocated room, seat, approved stay period, check-in, and check-out information.                                                  | Room Allocation                                                                                                                                                                                                     |
| Request stay extension/cancellation and report dormitory issues for staff action.                                                          | Hostel Manager                                                                                                                                                                                                      |
| Ask dormitory questions and request supported navigation or device assistance.                                                             | Hostel AI                                                                                                                                                                                                           |

**Collaborations:**

Dorm User --(uses identity from)--> User  
 Dorm User --(submits / tracks)--> Dormitory Application  
 Dorm User --(views)--> Room Allocation  
 Dorm User --(requests service from)--> Hostel Manager  
 Dorm User --(asks)--> Hostel AI

**Table CC.24: CRC Card for Dormitory Application**

| Class Name: Dormitory Application                                                                                                                                                                                                                    |                                                                                                                                                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Attributes                                                                                                                                                                                                                                           | Methods                                                                                                                                            |
| -applicationID<br>-dormUserID<br>-nidNumber<br>-preferredAccommodation<br>-expectedStart<br>-expectedEnd<br>-seatType<br>-reason<br>-applicationStatus<br>-currentApprovalStage<br>-approvalHistory<br>-feeAmount<br>-waiverStatus<br>-paymentStatus | +submit()<br>+updateStatus()<br>+recordApproval()<br>+calculateFee()<br>+applyWaiver()<br>+markPendingForAvailability()<br>+confirmPaymentStatus() |
| Responsibility                                                                                                                                                                                                                                       | Collaboration                                                                                                                                      |
| Store the accommodation request, preferences, expected stay, and required participant information.                                                                                                                                                   | Dorm User                                                                                                                                          |
| Move through the required approval sequence and retain each authority decision in approval history.                                                                                                                                                  | Approval Authority                                                                                                                                 |
| Allow the manager to review availability and complete final dormitory processing.                                                                                                                                                                    | Hostel Manager                                                                                                                                     |
| Send the determined fee/waiver for processing and receive payment confirmation.                                                                                                                                                                      | Payment System                                                                                                                                     |
| Provide the approved and paid application used to create the room/seat allocation.                                                                                                                                                                   | Room Allocation                                                                                                                                    |

**Collaborations:**

Dormitory Application --(belongs to)--> Dorm User  
 Dormitory Application --(is reviewed by)--> Approval Authority  
 Dormitory Application --(is finalized by)--> Hostel Manager  
 Dormitory Application --(requests payment from)--> Payment System  
 Dormitory Application --(creates allocation for)--> Room Allocation

**Table CC.25: CRC Card for Hostel Manager**

| <b>Class Name: Hostel Manager</b>                                                        |                                                                                                                                                                                                      |
|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Attributes</b>                                                                        | <b>Methods</b>                                                                                                                                                                                       |
| -managerID<br>-userID<br>-permissions<br>-occupancySummary                               | +reviewApplication()<br>+checkAvailability()<br>+allocateRoom()<br>+recordCheckIn()<br>+recordCheckOut()<br>+manageOccupancy()<br>+approveStayChange()<br>+manageIssue()<br>+generateMonthlyReport() |
| <b>Responsibility</b>                                                                    | <b>Collaboration</b>                                                                                                                                                                                 |
| Review the application after/between required approvals and manage pending availability. | Dormitory Application                                                                                                                                                                                |
| Record check-in/check-out, process stay changes, and manage reported dormitory issues.   | Dorm User                                                                                                                                                                                            |
| Check room-level capacity and availability while processing accommodation.               | Room                                                                                                                                                                                                 |
| Select an appropriate available seat and release it when the stay ends.                  | Seat                                                                                                                                                                                                 |
| Create/manage allocations and keep occupancy information current.                        | Room Allocation                                                                                                                                                                                      |
| Receive the formal approval outcome before completing final allocation decisions.        | Approval Authority                                                                                                                                                                                   |

**Collaborations:**

Hostel Manager --(reviews)--> Dormitory Application  
 Hostel Manager --(manages stay / issues for)--> Dorm User  
 Hostel Manager --(checks)--> Room  
 Hostel Manager --(selects)--> Seat  
 Hostel Manager --(creates / manages)--> Room Allocation  
 Hostel Manager --(receives outcome from)--> Approval Authority

**Table CC.26: CRC Card for Approval Authority**

| <b>Class Name: Approval Authority</b>                                                          |                                                                                              |
|------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| <b>Attributes</b>                                                                              | <b>Methods</b>                                                                               |
| -authorityID<br>-userID<br>-authorityRole<br>-sequenceOrder<br>-decisionStatus                 | +reviewApplication()<br>+approve()<br>+reject()<br>+addRemark()<br>+forwardToNextAuthority() |
| <b>Responsibility</b>                                                                          | <b>Collaboration</b>                                                                         |
| Review the dormitory application, add remarks, and record approve/reject decisions.            | Dormitory Application                                                                        |
| Forward the completed approval flow to the manager for availability and allocation processing. | Hostel Manager                                                                               |
| Use emailed direct approval links and approval notifications when required.                    | Email Service                                                                                |

**Collaborations:**

Approval Authority --(approves / rejects)--> Dormitory Application  
Approval Authority --(forwards outcome to)--> Hostel Manager  
Approval Authority --(sends approval link through)--> Email Service

**Table CC.27: CRC Card for Room**

| <b>Class Name: Room</b>                                                                                   |                                                                               |
|-----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| <b>Attributes</b>                                                                                         | <b>Methods</b>                                                                |
| -roomID<br>-roomNumber<br>-roomType<br>-totalCapacity<br>-occupiedSeats<br>-availableSeats<br>-roomStatus | +viewAvailability()<br>+reserveSeat()<br>+releaseSeat()<br>+updateOccupancy() |
| <b>Responsibility</b>                                                                                     | <b>Collaboration</b>                                                          |
| Aggregate its seats to maintain total, occupied, and available capacity.                                  | Seat                                                                          |
| Provide room type, capacity, and availability information to dormitory management.                        | Hostel Manager                                                                |
| Update occupancy when an allocation reserves or releases a seat in the room.                              | Room Allocation                                                               |

**Collaborations:**

Room --(contains)--> Seat  
 Room --(provides availability to)--> Hostel Manager  
 Room --(has occupancy updated by)--> Room Allocation

**Table CC.28: CRC Card for Seat**

| <b>Class Name: Seat</b>                                                                   |                                                   |
|-------------------------------------------------------------------------------------------|---------------------------------------------------|
| <b>Attributes</b>                                                                         | <b>Methods</b>                                    |
| -seatID<br>-roomID<br>-seatNumber<br>-seatType<br>-availabilityStatus                     | +checkAvailability()<br>+allocate()<br>+release() |
| <b>Responsibility</b>                                                                     | <b>Collaboration</b>                              |
| Belong to a room and contribute to that room's capacity and availability.                 | Room                                              |
| Become linked to a participant through an approved room allocation and later be released. | Room Allocation                                   |
| Provide the manager with seat type and availability for allocation decisions.             | Hostel Manager                                    |

**Collaborations:**

Seat --(belongs to)--> Room  
 Seat --(is allocated / released through)--> Room Allocation  
 Seat --(is selected by)--> Hostel Manager

**Table CC.29: CRC Card for Room Allocation**

| <b>Class Name: Room Allocation</b>                                                                                                                                |                                                                                                           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| <b>Attributes</b>                                                                                                                                                 | <b>Methods</b>                                                                                            |
| -allocationID<br>-applicationID<br>-dormUserID<br>-roomID<br>-seatID<br>-allocationDate<br>-checkInDate<br>-checkOutDate<br>-approvedEndDate<br>-allocationStatus | +allocate()<br>+activateStay()<br>+updateStayDates()<br>+recordCheckOut()<br>+release()<br>+viewDetails() |
| <b>Responsibility</b>                                                                                                                                             | <b>Collaboration</b>                                                                                      |
| Create the allocation only from an approved application with satisfied payment/waiver conditions.                                                                 | Dormitory Application                                                                                     |
| Provide the participant with allocated room, seat, stay, check-in, and check-out details.                                                                         | Dorm User                                                                                                 |
| Update the selected room's occupancy when accommodation begins or ends.                                                                                           | Room                                                                                                      |
| Mark the selected seat allocated and later available after checkout/cancellation.                                                                                 | Seat                                                                                                      |
| Allow the manager to allocate, check in, check out, and maintain the active stay.                                                                                 | Hostel Manager                                                                                            |

**Collaborations:**

Room Allocation --(is based on)--> Dormitory Application  
 Room Allocation --(assigns room / seat to)--> Dorm User  
 Room Allocation --(updates occupancy of)--> Room  
 Room Allocation --(allocates / releases)--> Seat  
 Room Allocation --(is managed by)--> Hostel Manager

**Table CC.30: CRC Card for Hostel AI**

| Class Name: Hostel AI                                                                              |                                                                                                                                |
|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Attributes                                                                                         | Methods                                                                                                                        |
| -assistantID<br>-userQuery<br>-roomContext<br>-generatedResponse<br>-actionStatus<br>-mapReference | +answerDormitoryQuestion()<br>+retrieveRoomInfo()<br>+provideNavigation()<br>+controlSupportedDevice()<br>+escalateToManager() |
| Responsibility                                                                                     | Collaboration                                                                                                                  |
| Receive dormitory questions and supported navigation/device requests from the participant.         | Dorm User                                                                                                                      |
| Use authorized room-specific information such as Wi-Fi details or map/location guidance.           | Room                                                                                                                           |
| Escalate unsupported questions, judgments, or service problems to dormitory management.            | Hostel Manager                                                                                                                 |
| Use the common AI capability to interpret requests and generate dormitory responses.               | AI Service                                                                                                                     |

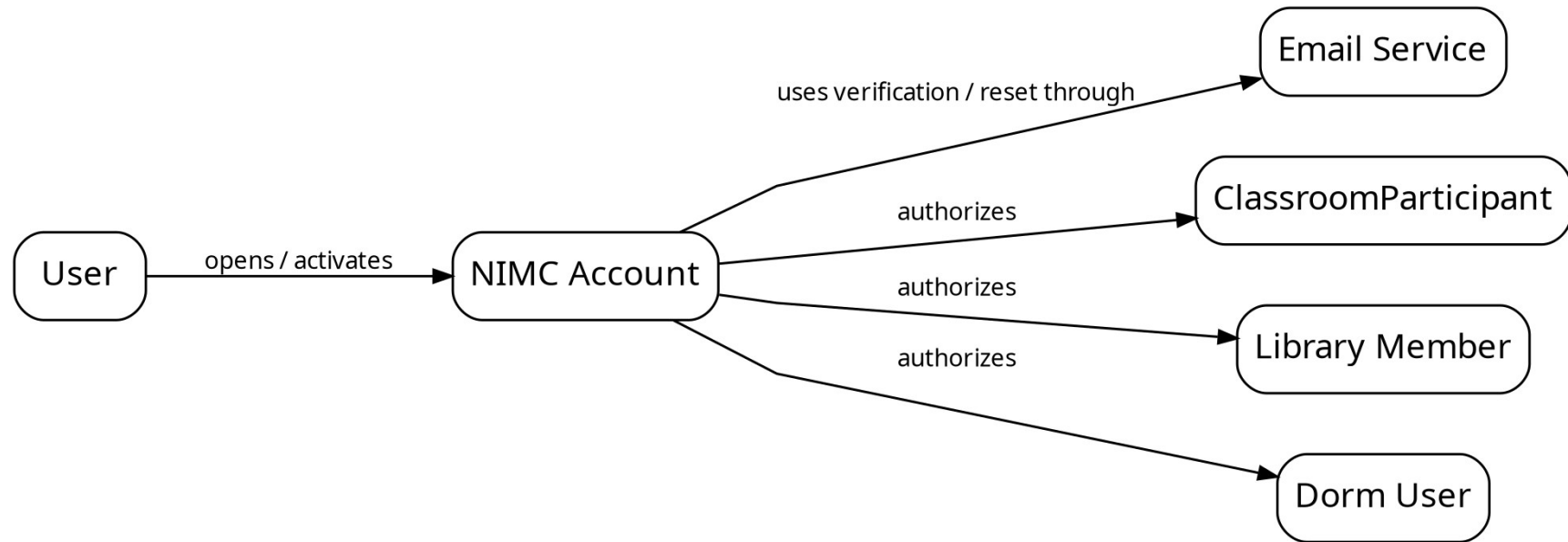
**Collaborations:**

Hostel AI --(assists)--> Dorm User  
 Hostel AI --(retrieves room info from)--> Room  
 Hostel AI --(escalates to)--> Hostel Manager  
 Hostel AI --(calls)--> AI Service

## 6. Class Responsibility Collaboration Diagrams

Only four CRC collaboration diagrams are used: Authentication System, Classroom, Library, and Dormitory. Each arrow label states the collaboration action between the two classes.

### 6.1 Authentication System



*Fig CRC.1: Authentication System Collaboration Diagram*

6.2 Classroom Module

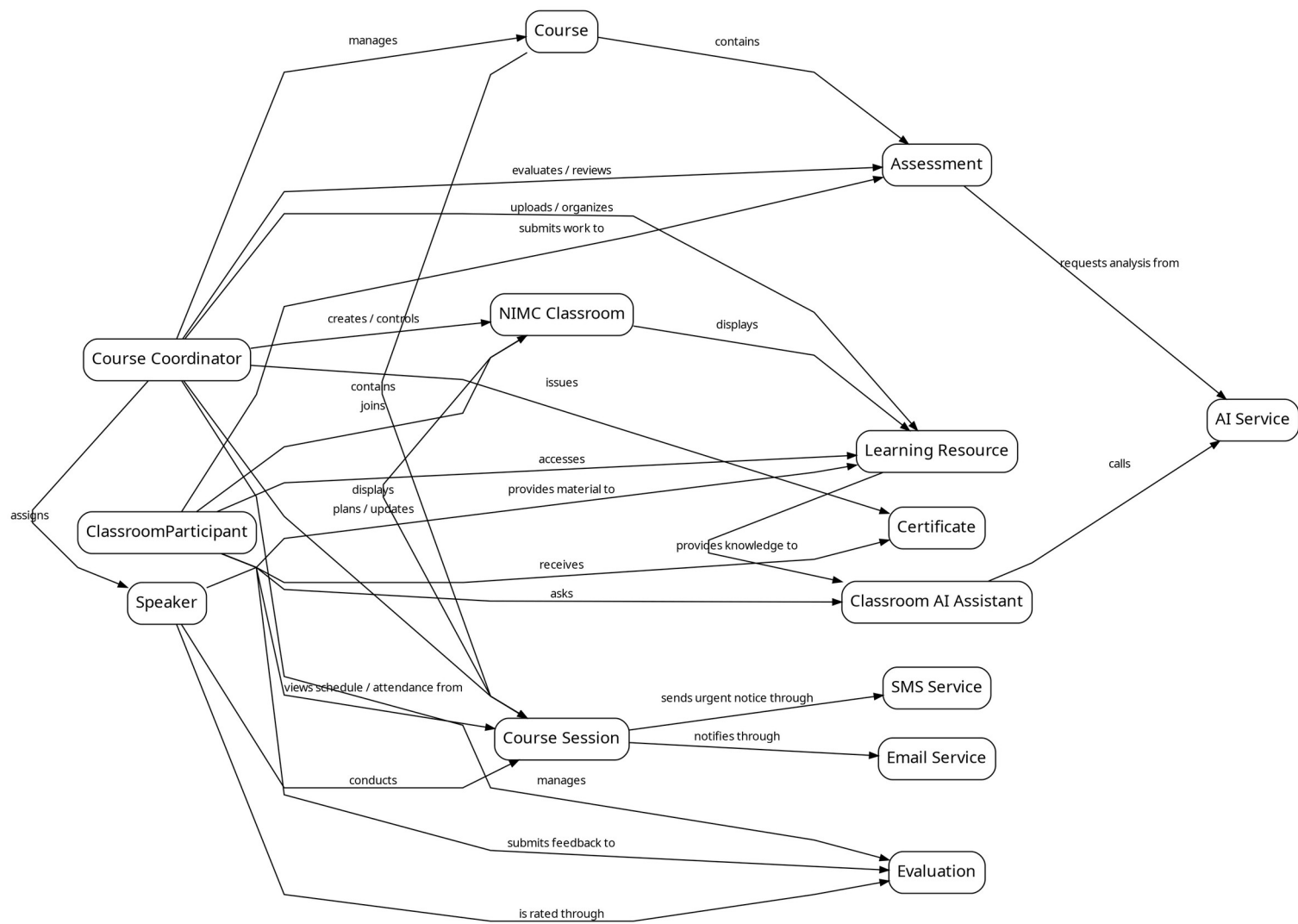


Fig CRC.2: Classroom Module Collaboration Diagram

### 6.3 Library Module

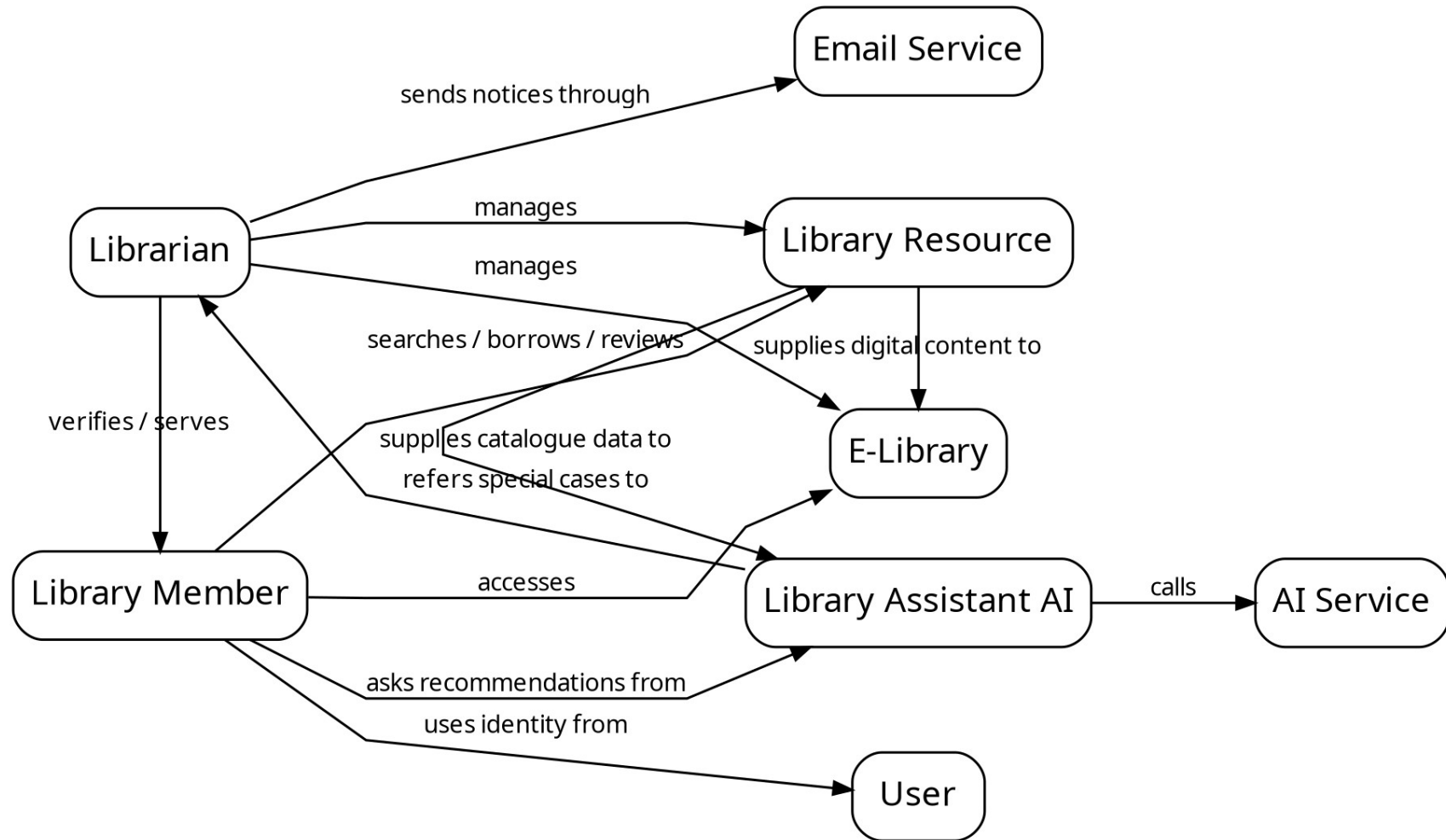


Fig CRC.3: Library Module Collaboration Diagram

## 6.4 Dormitory Module

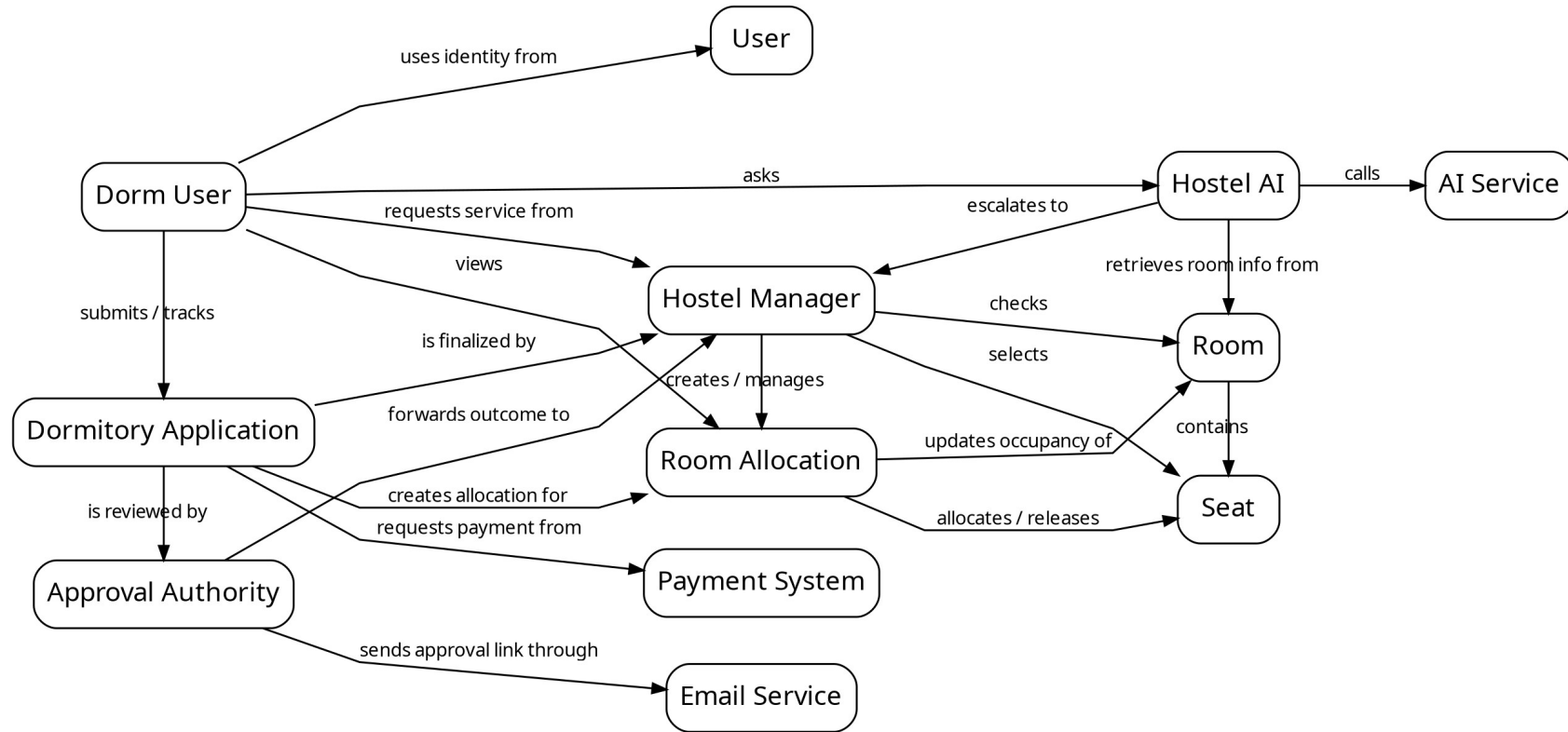


Fig CRC.4: Dormitory Module Collaboration Diagram